SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA0604
COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED
CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15
Total Marks: 25

Annexure A
APPLICATION BASED QUESTIONS

Code of Conduct:
I **SHUNMUGA PRIYA G (192412051)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
SHUNMUGA PRIYA G

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

SOLUTION:

Conceptual Understanding

- Box A: 6 red, 4 blue, 2 green balls. Total = 12.
- Box B: 5 red, 7 blue, 3 green balls. Total = 15.
- Box C: 8 red, 2 blue, 5 green balls. Total = 15

Box B is twice as likely to be selected as Box A or Box C.

Step wise calculation:
Step 1: Calculate the probability of selecting each box

The probability ratio is 1:2:1.

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Accuracy of answer

Step 2: Calculate the probability of drawing a blue ball from each box

$$P(Blue|A) = \frac{4}{12} = \frac{1}{3}$$

$$P(Blue|B) = \frac{7}{15}$$

$$P(Blue|C) = \frac{2}{15}$$

Appliaction of knowledge

Step 3: Apply Bayes' theorem

Bayes' theorem is

$$P(C|Blue) = \frac{P(Blue|C)P(C)}{P(Blue)}$$

Using the law of total probability,

$$P(Blue) = P(Blue|A)P(A) + P(Blue|B)P(B) + P(Blue|C)P(C)$$

Substituting the values,

$$P(Blue) = \frac{1}{3} \times \frac{1}{4} + \frac{7}{15} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}$$

$$P(Blue) = \frac{1}{12} + \frac{7}{30} + \frac{1}{30}$$

$$P(Blue) = \frac{7}{20}$$

Clarity of Presentation

Therefore,

$$P(C|Blue) = \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{7}{20}}$$

$$= \frac{1/30}{7/20} = \frac{20}{210} = \frac{2}{21}$$

Final Answer

P(C|blue)=2/12=9.52%

The probability that the selected box was Box C, given that a blue ball was drawn, is **9.52%**.

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2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

SOLUTION:

Conceptual Understanding

Given Data

Section	Red Cars	Blue Cars	Black Cars	Total Cars
A	**5**	**3**	**2**	**10**
B	**4**	**6**	**5**	**15**
C	**7**	**2**	**6**	**15**

Section B is twice as likely to be selected as Section A or Section C.

Step 1: Calculate the prior probabilities

The selection ratio is 1: 2: 1.

Therefore,

$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$

Accuracy of answer

Step 2: Calculate the probability of selecting a blue car

For Section A

$P(Blue|A) = \frac{3}{10}$

For Section B,

$P(Blue|B) = \frac{6}{15} = \frac{2}{5}$

For Section C,

$P(Blue|C) = \frac{2}{15}$

Appliaction of knowledge

Step 3: Apply Bayes' theorem

We need to determine

$P(C|Blue) = \frac{P(Blue|C)P(C)}{P(Blue)}$

Using the law of total probability**,**

$P(Blue) = \frac{3}{10} \times \frac{1}{4} + \frac{2}{5} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}$

$P(Blue) = \frac{3}{40} + \frac{1}{5} + \frac{1}{30}$

Taking the LCM of 40, 5, and 30:

$P(Blue) = \frac{9}{120} + \frac{24}{120} + \frac{4}{120}$

$P(Blue) = \frac{37}{120}$

Clarity of Presentation

Therefore,

$$P(C|Blue) = \frac{\frac{2}{15}\times\frac{1}{4}}{\frac{37}{120}}$$

$$= \frac{1}{30} \times \frac{120}{37}$$

$$= \frac{4}{37}$$

P(C|blue)=4/37=10.81%

The probability that the blue car came from Section C is **10.81%.**

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3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

SOLUTION:

Conceptual Understanding

Given Data

- Malignant samples = 1800
- Benign samples = 1800
- Total dataset = 3600 samples
- Testing data = 20%

- True Positives (TP) = 210
- True Negatives (TN) = 190
- False Positives (FP) = 30

We must calculate the False Negatives (FN), Accuracy, Precision, Sensitivity, and Specificity.

Step 1: Calculate the total number of testing samples

$Testingsamples = 3600 \times \frac{20}{100}$

Testing samples=720

Step 2: Calculate False Negatives

The total number of testing samples is

$TP + TN + FP + FN = 720$

Substituting the given values,

$210 + 190 + 30 + FN = 720$

$FN = 720 - 430$

FN=290

Accuracy of answer

Step 3: Calculate Accuracy

Accuracy measures the proportion of all predictions that are correct.

$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \times 100$

$= \frac{210+190}{720} \times 100$

Accuracy =55.56%

Step 4: Calculate Precision

Precision measures the proportion of predicted malignant cases that are actually malignant.

$$Precision = \frac{TP}{TP+FP} \times 100$$
$$= \frac{210}{210+30} \times 100$$

Precision=87.5%

Appliaction of knowledge

Step 5: Calculate Sensitivity (Recall)

Sensitivity measures the proportion of actual malignant cases correctly identified by the model.

$$Sensitivity = \frac{TP}{TP+FN} \times 100$$

$$= \frac{210}{210+290} \times 100$$

Sensitivity=42%

Step 6: Calculate Specificity

Specificity measures the proportion of actual benign cases correctly identified by the model.

$$Specificity = \frac{TN}{TN+FP} \times 100$$

$$= \frac{190}{190+30} \times 100$$

Specificity=86.36%

Clarity of Presentation

Effectiveness in Medical Diagnosis

The model has a precision of 87.5%, meaning that most cases it predicts as malignant are actually malignant. Its specificity of 86.36% indicates that it correctly identifies many benign cases.

However, its sensitivity is only 42%, meaning it detects only 42% of the actual malignant cases. It misses 58% of them, which is a serious concern in breast cancer screening because missed cancer cases can delay diagnosis and treatment.

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4. Examine the performance of a machine-learning model using its training and

validation loss values over multiple epochs:

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4

(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2

(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0

(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1

(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3

(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6

(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9

(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.

ii) Suggest two effective techniques to reduce overfitting in this model.

iii) Discuss how overfitting affects the model's generalization ability on unseen data

SOLUTION:

Conceptual Understanding

Given Data

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

Accuracy of answer

(i) Identify the epoch at which the model begins to overfit

Overfitting occurs when the model continues to improve on the training data but performs increasingly poorly on unseen data.

- Epochs 1–3: Both training and validation loss decrease.
- Epoch 4: Training loss decreases to 2.0, but validation loss increases from 2.0 to 2.1.
- Epochs 5–8: Training loss continues decreasing, while validation loss continues increasing.

The model begins to overfit at Epoch 4.

Application of Knowledge

(ii) Suggest two effective techniques to reduce overfitting

1. Regularization (L1 or L2)

Regularization adds a penalty to the model's loss function for overly complex parameter values. It discourages the model from relying too heavily on particular training examples and can improve generalization.

2. Early Stopping

Early stopping monitors validation loss during training. Training is stopped when validation performance no longer improves. In this example, the model could retain the weights from Epoch 3 because that epoch has the lowest validation loss.

Clarity &Presentation

(iii) Discuss how overfitting affects generalization ability

Generalization is the ability of a trained model to make accurate predictions on new, unseen data.When overfitting occurs, the model learns not only meaningful patterns but also noise and specific details in the training dataset. Consequently, its training loss continues decreasing, while its validation loss increases.This indicates that the model is becoming better at fitting its training data but worse at predicting unseen data. **Overfitting reduces generalization and can be controlled through regularization, early stopping, dropout, or suitable data augmentation.**

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5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

(h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

iii) Suggest two methods to improve validation performance.

SOLUTION:

Conceptual Understanding

Given Data

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%

3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

Accuracy of answer

(i) Identify the epoch at which the model starts overfitting

- Epochs 1–4: Both training and validation accuracy increase.
- Epoch 5: Training accuracy increases to 80%, and validation accuracy reaches its highest value of 73%.
- Epoch 6: Training accuracy increases to 85%, but validation accuracy decreases to 72%.
- Epochs 7–8: Training accuracy continues increasing, while validation accuracy continues decreasing.

The model starts overfitting at Epoch 6.

Application of Knowledge

(ii) Explain why validation accuracy decreases despite training accuracy increasing

Training accuracy measures how well the model predicts the examples used during training. Validation accuracy measures how well it performs on separate data that was not used to update its parameters.

As training continues, the model may memorize specific training examples and learn noise rather than general patterns. This causes training accuracy to increase while its ability to predict unseen examples decreases.

The increasing gap between training and validation accuracy indicates overfitting and reduced generalization.

Clarity &Presentation

(iii) Suggest two methods to improve validation performance

1. Dropout

Dropout randomly deactivates a proportion of neurons during training. This prevents the model from depending too heavily on particular neurons and encourages it to learn more robust patterns.

2. Early Stopping

Early stopping monitors validation accuracy or validation loss and stops training when validation performance stops improving. Other possible methods include data augmentation, L2 regularization, and reducing model complexity.

The model's validation performance deteriorates after Epoch 5 because of overfitting. Using dropout and early stopping can help improve its performance on unseen data.
